

- Genetics -

⇒ Sexual reproduction → variations.

⇒ Pea plant → Pisum sativum

→ Short life span

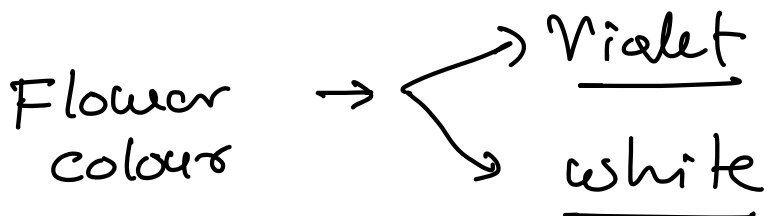
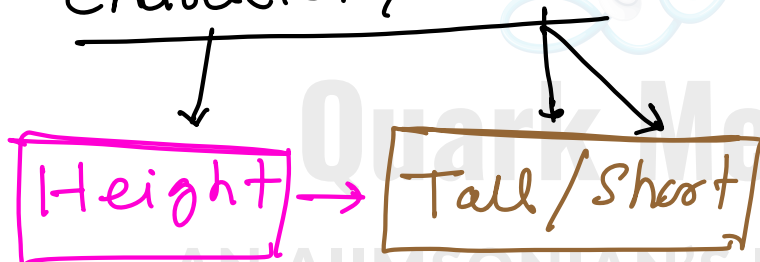
→ Inheritable variation

→ Characters with contrasting traits

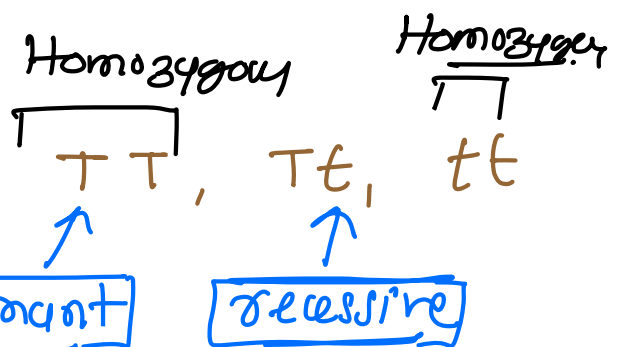
⇒ Factors ⇒ gene

⇒ 1856 - 1863 → 7 years

⇒ Character / Trait ⇒



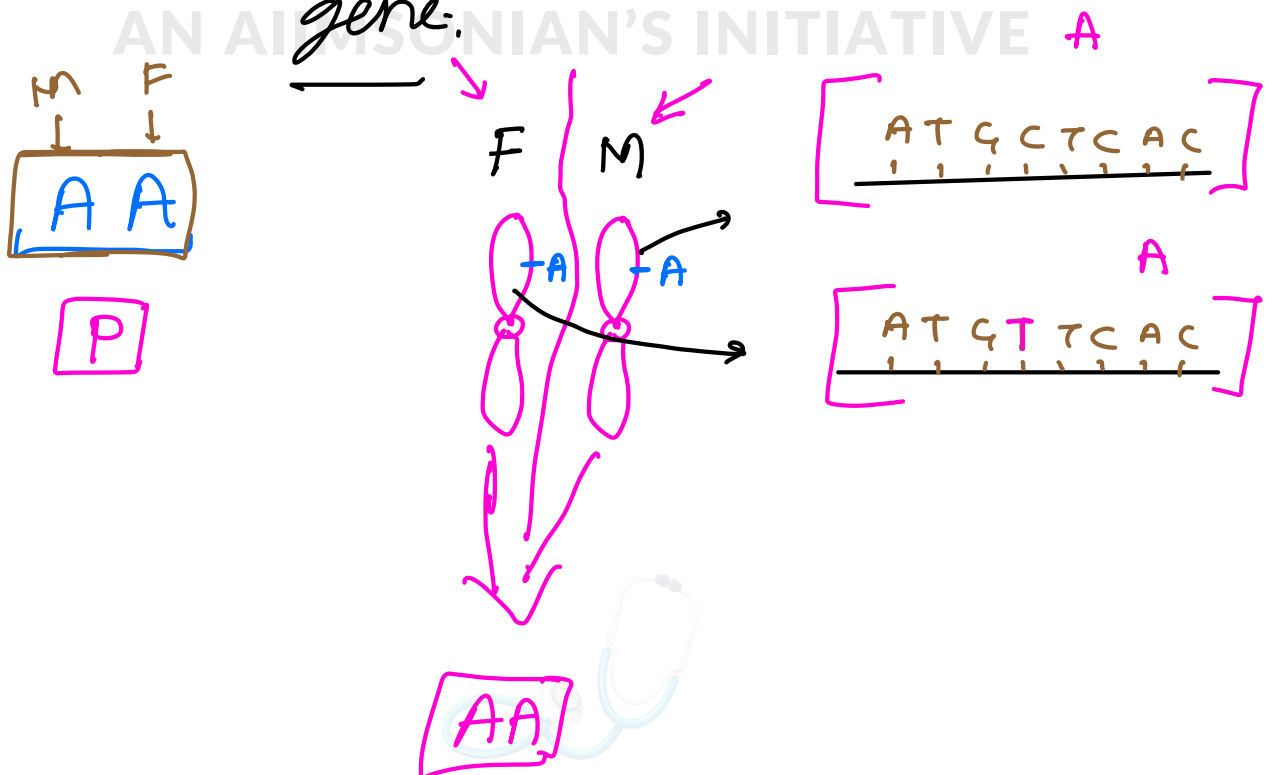
⇒ Factor occur in pairs



Tt → In a heterozygous pair the

Character that is expressed is called a Dominant character.
 But the ^{Factor} character that is not expressed at this stage is called recessive Factor.

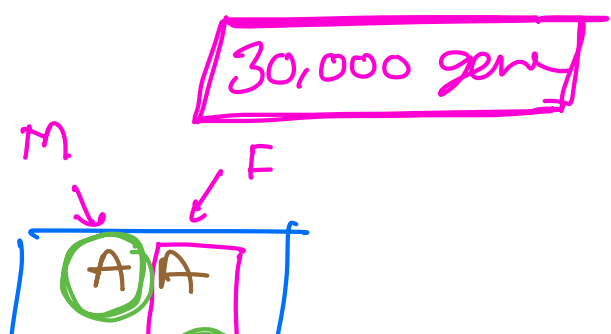
⇒ Allele → Slight modified form of a gene.

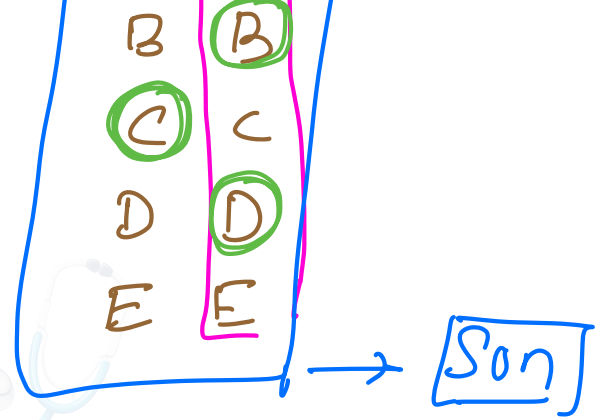


⇒ gene → Segment of DNA that codes for a particular character or a set of character.

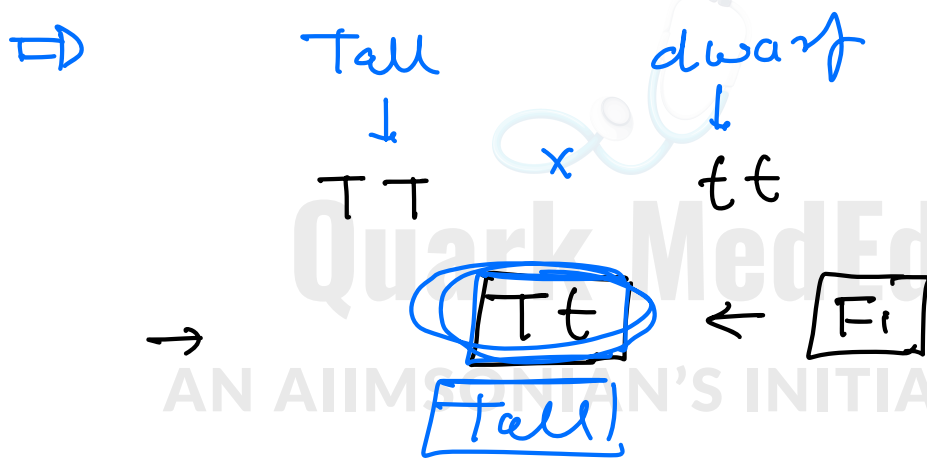
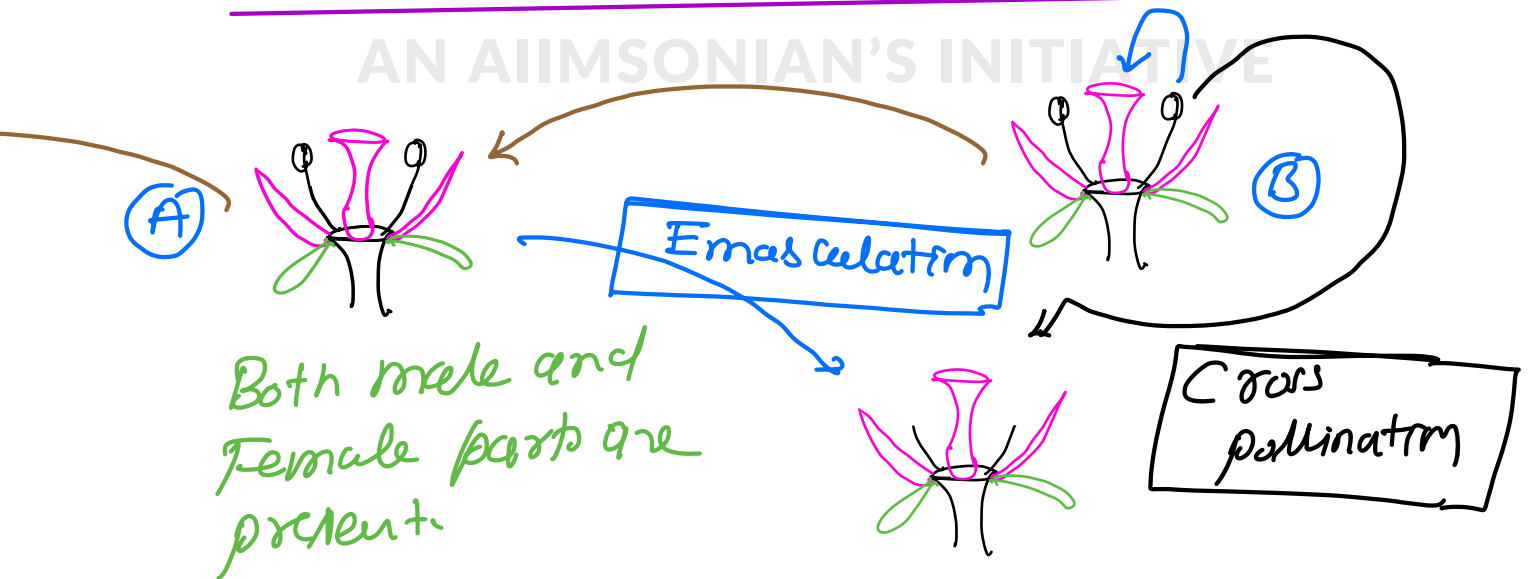
⇒ AA ⇒

gene ⇒



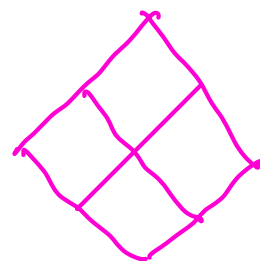


⇒ Mendel's Hybridization Experiment ⇒



$$Tt \times Tt$$

	T	t
T	TT	Tt
t	Tt	tt



Expected
Appearance
phenotypic
Ratio

⇒ Tall : dwarf
3 : 1

Genotypic Ratio $\left[\begin{array}{c} TT : Tt : tt \\ 1 : 2 : 1 \end{array} \right]$

$\Rightarrow$ Pure Lines $\Rightarrow$

$\uparrow$

they show stable trait in heritage from generation to generation

Tall

x Dwarf

TT

tt

$\left\{ \begin{array}{l} Tt \times Tt \\ TT \times TT \\ TT \times TT \\ \downarrow \\ TT \end{array} \right.$

$\Rightarrow$ In Breeding depression $\Rightarrow$

Distal Family

Hb Hb

Hb^s Hb

Female

Close related

Hb^s Hb

male

Hb^s Hb, Hb Hb

Hb^s Hb^s, Hb^s Hb



Inbreeding
depression

⇒ Inbreeding depression be overcome
↓
by out crossing.

AN AIMSONIAN'S INITIATIVE

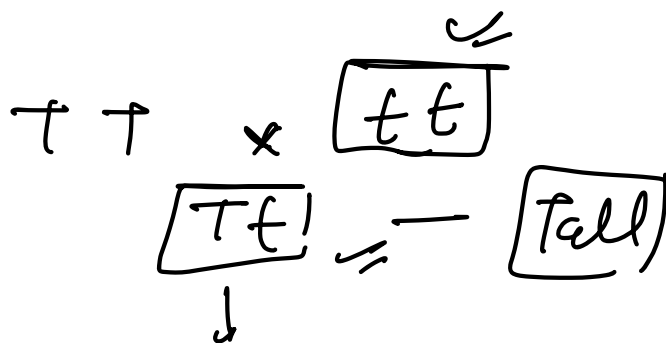
⇒ Law of Dominance →

- Factors occurs in pairs
- character are controlled by a pair of factor.
- In a dissimilar pair of factor (Heterozygous) the one that is expressed is called a dominant character and the one that is not expressed is called the recessive character.

AN AIMSONIAN'S INITIATIVE

⇒ Law of segregation →

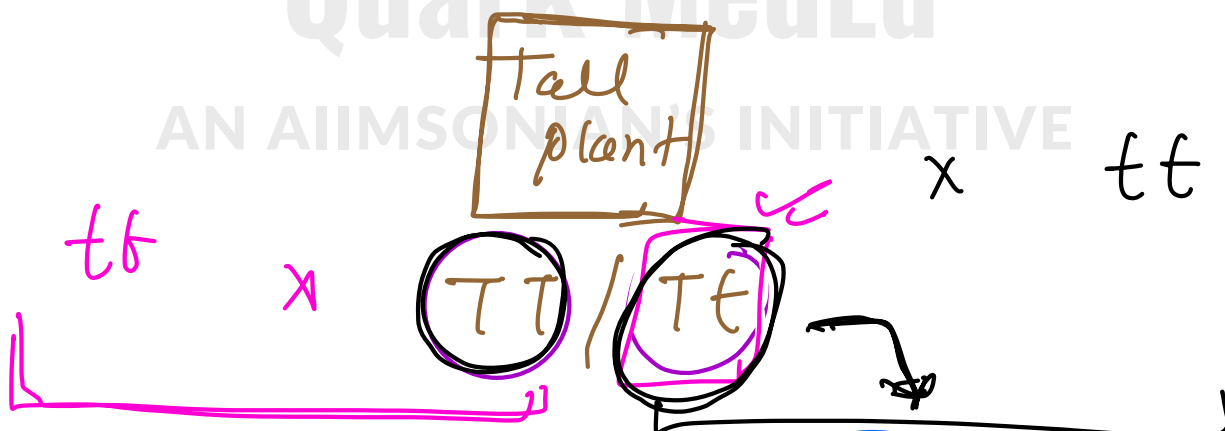
Alleles do not show any blending



$$Tt \times Tt$$

$$TT : Tt : tt \approx$$

⇒ Test Cross ⇒



	T	T
t	Tt	Tt
t	Tt	Tt

100% Tall

	T	t
t	Tt	tt
t	Tt	tt

50% Tall

Quark MedEd

AN AIIMSONIAN'S INITIATIVE